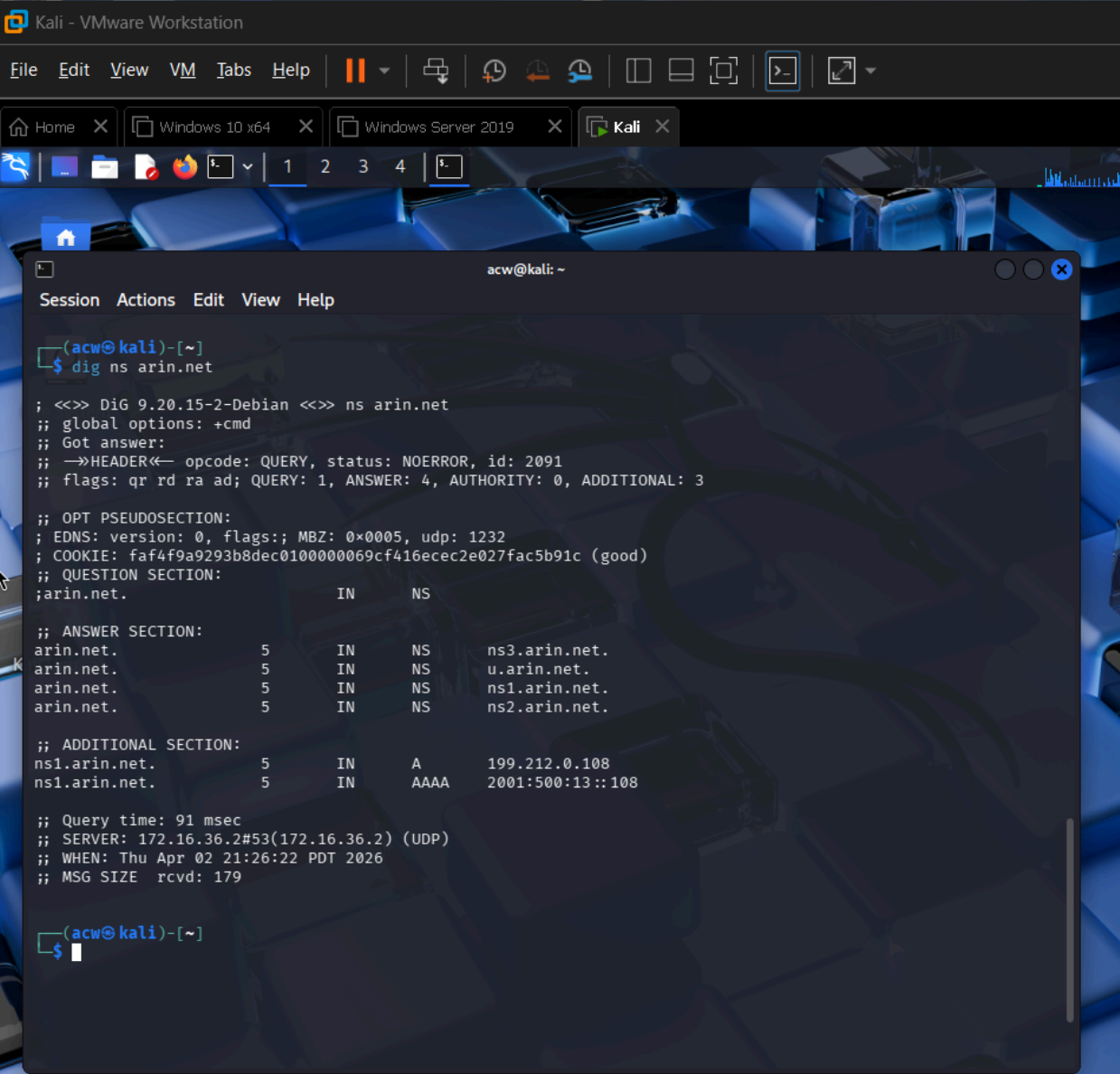


Lab Exercise 7.02: DNSSEC for Exploiting

1



```
Kali - VMware Workstation
File Edit View VM Tabs Help
Home x Windows 10 x64 x Windows Server 2019 x Kali x
1 2 3 4

acw@kali: ~
Session Actions Edit View Help

(acw@kali)-[~]
$ dig ns arin.net

; <<>> DiG 9.20.15-2-Debian <<>> ns arin.net
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 2091
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 3

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
; COOKIE: faf4f9a9293b8dec0100000069cf416ecec2e027fac5b91c (good)
;; QUESTION SECTION:
;arin.net.                IN      NS

;; ANSWER SECTION:
arin.net.                5       IN      NS      ns3.arin.net.
arin.net.                5       IN      NS      u.arin.net.
arin.net.                5       IN      NS      ns1.arin.net.
arin.net.                5       IN      NS      ns2.arin.net.

;; ADDITIONAL SECTION:
ns1.arin.net.           5       IN      A       199.212.0.108
ns1.arin.net.           5       IN      AAAA    2001:500:13::108

;; Query time: 91 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:26:22 PDT 2026
;; MSG SIZE rcvd: 179

(acw@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2

Kali - VMware Workstation

File Edit View VM Tabs Help

Home X Windows 10 x64 X Windows Server 2019 X Kali X

1 2 3 4

```
acw@kali: ~
Session Actions Edit View Help

;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 11451
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 3

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
; COOKIE: 806e4af40d70abdc0100000069cf41d81258672a8d9cab9 (good)
;; QUESTION SECTION:
;arin.net.                IN      NS

;; ANSWER SECTION:
arin.net.                  5       IN      NS      ns2.arin.net.
arin.net.                  5       IN      NS      ns3.arin.net.
arin.net.                  5       IN      NS      u.arin.net.
arin.net.                  5       IN      NS      ns1.arin.net.

;; ADDITIONAL SECTION:
ns1.arin.net.              5       IN      A        199.212.0.108
ns1.arin.net.              5       IN      AAAA     2001:500:13::108

;; Query time: 87 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:28:08 PDT 2026
;; MSG SIZE rcvd: 179

(acw@kali)-[~]
$ dig axfr @ns.arin.net arin.net
dig: couldn't get address for 'ns.arin.net': not found

(acw@kali)-[~]
$ dig axfr @ns1.arin.net arin.net

; <<>> DiG 9.20.15-2-Debian <<>> axfr @ns1.arin.net arin.net
; (2 servers found)
;; global options: +cmd
; Transfer failed.

(acw@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Kali - VMware Workstation

File Edit View VM Tabs Help

Home X Windows 10 x64 X Windows Server 2019 X Kali X

1 2 3 4

```
acw@kali: ~
Session Actions Edit View Help
(acw@kali)-[~]
$ dig axfr @ns.arin.net arin.net
dig: couldn't get address for 'ns.arin.net': not found

(acw@kali)-[~]
$ dig axfr @ns1.arin.net arin.net

; <<>> DiG 9.20.15-2-Debian <<>> axfr @ns1.arin.net arin.net
; (2 servers found)
;; global options: +cmd
; Transfer failed.

(acw@kali)-[~]
$ dig rit.arin.net

; <<>> DiG 9.20.15-2-Debian <<>> rit.arin.net
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 33673
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
; COOKIE: dd275e9575e352af0100000069cf4240edc3141b77d76319 (good)
;; QUESTION SECTION:
;rit.arin.net.                IN      A

;; AUTHORITY SECTION:
arin.net.                    5       IN      SOA     ns1.arin.net. bind.arin.net. 2021113585 10800 600 1209600 36
00

;; Query time: 291 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:29:52 PDT 2026
;; MSG SIZE rcvd: 114

(acw@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Kali - VMware Workstation

File Edit View VM Tabs Help

Home X Windows 10 x64 X Windows Server 2019 X Kali X

1 2 3 4

```
acw@kali: ~
Session Actions Edit View Help
00

;; Query time: 291 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:29:52 PDT 2026
;; MSG SIZE rcvd: 114

(acw@kali)-[~]
$ dig +dnssec rit.arin.net

; <<>> DiG 9.20.15-2-Debian <<>> +dnssec rit.arin.net
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NXDOMAIN, id: 29466
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 0

;; QUESTION SECTION:
;rit.arin.net.                IN      A

;; AUTHORITY SECTION:
arin.net.      5      IN      SOA      ns1.arin.net. bind.arin.net. 2021113585 10800 600 1209600 36
00
arin.net.      5      IN      RRSIG    SOA 8 2 43200 20260416120020 20260402110020 41655 arin.net.
UPVbSOC8lCuHc2zoc8UAfRyQ79zOy1qmM4oLNvipeQeYct1cqH22ZEJ +pWSYH7twLh3PSYLuvTHqCCGa0ch5LD5id64qqXANVJclCI/f3b
hkuAj M376SAPRUX54EyfxBFsQkwE3yq2/FA5WT1q1axRo0NrU8KEECJt0cs2 C1k=
arin.net.      5      IN      RRSIG    NSEC 8 2 3600 20260416120020 20260402110020 41655 arin.net.
CZN1Wvy4TqwD2R7YNPU+AarMgt9GaAfxrS3EsI+Xv9KKjmTAmuaG16dA Nmkk1kC4qAspU4knKwnt/vV70abgMonVMzjBduMdF1mJABNi6iI
J2qYw 0hkjdktRwD+M7/XarqychWaJsi6mDWLuZKaRtQ3/xn9Z/WC+E5YlkyfJ Ang=
arin.net.      5      IN      NSEC     _dmarc.arin.net. A NS SOA MX TXT AAAA RRSIG NSEC DNSKEY CAA

;; Query time: 219 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:30:41 PDT 2026
;; MSG SIZE rcvd: 452

(acw@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Home Windows 10 x64 Windows Server 2019 Kali
acw@kali: ~
Session Actions Edit View Help
arin.net. 5 IN NSEC _dmarc.arin.net. A NS SOA MX TXT AAAA RRSIG NSEC DNSKEY CAA
;; Query time: 219 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:30:41 PDT 2026
;; MSG SIZE rcvd: 452
(acw@kali)-[~]
$ dig +dnssec jonathan.arin.net
; <<>> Dig 9.20.15-2-Debian <<>> +dnssec jonathan.arin.net
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NXDOMAIN, id: 21946
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 0
;; QUESTION SECTION:
;jonathan.arin.net. IN A
;; AUTHORITY SECTION:
arin.net. 5 IN SOA ns1.arin.net. bind.arin.net. 2021113585 10800 600 1209600 3600
arin.net. 5 IN RRSIG SOA 8 2 43200 20260416120020 20260402110020 41655 arin.net.
UPVbSOC8lCuHc2zoc8UAfRyQ79zOy1qmM4oLNvipeQeYct1cqH22ZEJ +pWSYH7twLh3PSYLuvTHqCCGa0ch5LD5id64qqXANVJclCI/f3b
hkuAj M376SAPRUX54EyfxBFesQkwE3yq2/FA5WT1q1axRo0NrU8KEECJt0cs2 C1k=
arin.net. 5 IN RRSIG NSEC 8 2 3600 20260416120020 20260402110020 41655 arin.net.
CZN1Wvy4TqwD2R7YNPU+AarMgt9GaAfxrS3EsI+Xv9KKjmTAmuaG16dA Nmkk1kC4qAspU4knKwnt/vV70abgMonVMzjBduMdF1mJABNi6iI
J2qYw 0hkjdktRwD+M7/XarqychWaJsi6mDWLuZKaRtQ3/xn9Z/WC+E5YlkyfJ Ang=
arin.net. 5 IN NSEC _dmarc.arin.net. A NS SOA MX TXT AAAA RRSIG NSEC DNSKEY CAA
;; Query time: 47 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:32:29 PDT 2026
;; MSG SIZE rcvd: 457
(acw@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Kali - VMware Workstation

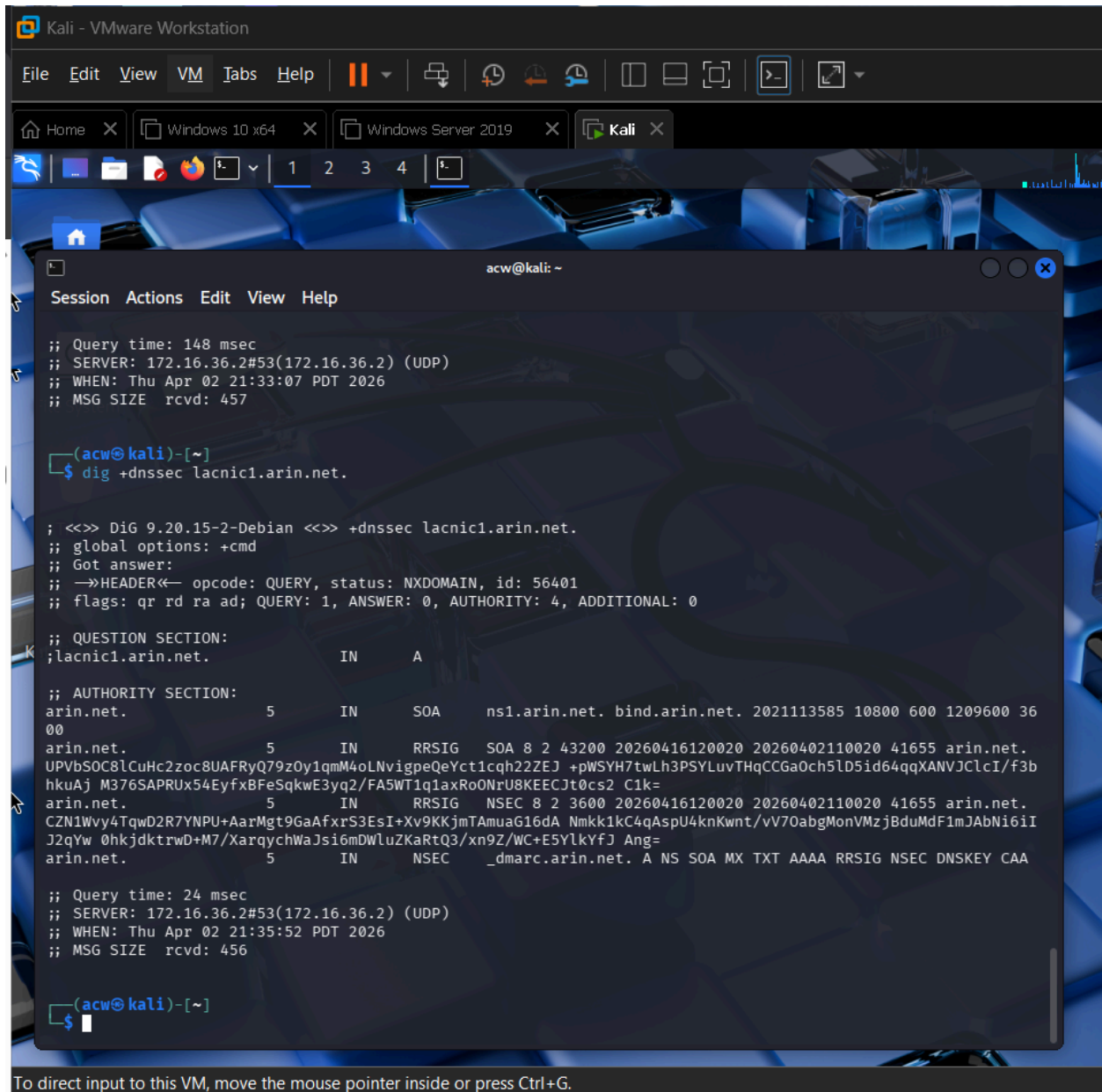
File Edit View VM Tabs Help

Home X Windows 10 x64 X Windows Server 2019 X Kali X

Minimize all open windows and show the desktop

```
acw@kali: ~  
Session Actions Edit View Help  
arin.net.          5      IN      NSEC      _dmarc.arin.net. A NS SOA MX TXT AAAA RRSIG NSEC DNSKEY CAA  
  
;; Query time: 47 msec  
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)  
;; WHEN: Thu Apr 02 21:32:29 PDT 2026  
;; MSG SIZE rcvd: 457  
  
(acw@kali)-[~]  
$ dig +dnssec kim-can1.arin.net  
  
; <<>> Dig 9.20.15-2-Debian <<>> +dnssec kim-can1.arin.net  
;; global options: +cmd  
;; Got answer:  
;; -->HEADER<-- opcode: QUERY, status: NXDOMAIN, id: 42941  
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 0  
  
;; QUESTION SECTION:  
;kim-can1.arin.net.          IN      A  
  
;; AUTHORITY SECTION:  
arin.net.          5      IN      SOA      ns1.arin.net. bind.arin.net. 2021113585 10800 600 1209600 3600  
arin.net.          5      IN      RRSIG    SOA 8 2 43200 20260416120020 20260402110020 41655 arin.net.  
UPVbSOC8lCuHc2zoc8UAFryQ79zOy1qmM4oLNvipeQeYct1cqh22ZEJ +pWSYH7twLh3PSYLuvTHqCCGa0ch5LD5id64qqXANVJclC/f3b  
hkuAj M376SAPRUX54EyfxBFesQkwE3yq2/FA5WT1q1axRo0NrU8KEECJt0cs2 C1k=  
arin.net.          5      IN      RRSIG    NSEC 8 2 3600 20260416120020 20260402110020 41655 arin.net.  
CZN1Wvy4TqwD2R7YNPU+AarMgt9GaAfxrS3EsI+Xv9KKjmTAmuaG16dA Nmkk1kC4qAspU4knKwnt/vV70abgMonVMzjBduMdF1mJABNi6iI  
J2qYw 0hkjdktwrD+M7/XarqychWaJsi6mDWLuZKaRtQ3/xn9Z/WC+E5YlkyfJ Ang=  
arin.net.          5      IN      NSEC      _dmarc.arin.net. A NS SOA MX TXT AAAA RRSIG NSEC DNSKEY CAA  
  
;; Query time: 148 msec  
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)  
;; WHEN: Thu Apr 02 21:33:07 PDT 2026  
;; MSG SIZE rcvd: 457  
  
(acw@kali)-[~]  
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
acw@kali: ~  
Session Actions Edit View Help  
;; Query time: 148 msec  
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)  
;; WHEN: Thu Apr 02 21:33:07 PDT 2026  
;; MSG SIZE rcvd: 457  
  
(acw@kali)-[~]  
$ dig +dnssec lacnic1.arin.net.  
  
; <<>> DiG 9.20.15-2-Debian <<>> +dnssec lacnic1.arin.net.  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 56401  
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 0  
  
;; QUESTION SECTION:  
;lacnic1.arin.net. IN A  
  
;; AUTHORITY SECTION:  
arin.net. 5 IN SOA ns1.arin.net. bind.arin.net. 2021113585 10800 600 1209600 3600  
arin.net. 5 IN RRSIG SOA 8 2 43200 20260416120020 20260402110020 41655 arin.net.  
UPVbSOC8lCuHc2zoc8UAFryQ79zOy1qmM4oLNvipeQeYct1cqh22ZEJ +pWSYH7twLh3PSYLuvTHqCCGa0ch5LD5id64qqXANVJclCI/f3b  
hkuAj M376SAPRux54EyfxBFesqkwE3yq2/FA5WT1q1axRoONrU8KEECJt0cs2 C1k=  
arin.net. 5 IN RRSIG NSEC 8 2 3600 20260416120020 20260402110020 41655 arin.net.  
CZN1Wvy4TqwD2R7YNPU+AarMgt9GaAfxrS3EsI+Xv9KKjmTAmuaG16dA Nmkk1kC4qAspU4knKwnt/vV70abgMonVMzjBduMdF1mJABNi6iI  
J2qYw 0hkjdktwD+M7/XarqychWaJsi6mDWLuZKaRtQ3/xn9Z/WC+E5YlkyfJ Ang=  
arin.net. 5 IN NSEC _dmarc.arin.net. A NS SOA MX TXT AAAA RRSIG NSEC DNSKEY CAA  
  
;; Query time: 24 msec  
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)  
;; WHEN: Thu Apr 02 21:35:52 PDT 2026  
;; MSG SIZE rcvd: 456  
  
(acw@kali)-[~]  
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

What would be the next resource record after lightning.arin.net?

According to the DNSSEC output in the terminal, the next resource record after lightning.arin.net in the canonical chain is _dmarc.arin.net

```
Kali - VMware Workstation
File Edit View VM Tabs Help
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acw@kali: ~
Session Actions Edit View Help
arin.net. 5 IN NSEC _dmarc.arin.net. A NS SOA MX TXT AAAA RRSIG NSEC DNSKEY CAA

;; Query time: 24 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:35:52 PDT 2026
;; MSG SIZE rcvd: 456

(acw@kali)-[~]
$ dig +dnssec rit.icann.org.

; <<>> DiG 9.20.15-2-Debian <<>> +dnssec rit.icann.org.
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 10430
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 0

;; QUESTION SECTION:
;rit.icann.org. IN A

;; AUTHORITY SECTION:
icann.org. 5 IN SOA sns.dns.icann.org. noc.dns.icann.org. 2026040207 10800 3600
1209600 3600
icann.org. 5 IN RRSIG SOA 13 2 3600 20260424091021 20260403014211 54015 icann.org.
j4E73oCUDysYlXoNVjWN5hNFJby+ErRZ4pskbzL9/WKBG0PK01BVEVvM ePPWYqG+Kke5Pdhx8HyvWqqsDYxliA=
9ldamu1tist5dbb9qtkntegrasihk0af.icann.org. 5 IN RRSIG NSEC3 13 3 3600 20260416004615 20260325231159 54015 i
cann.org. wJh2U17Id+GaUrWG+p5xgngwR/O/cFjy46v8uodhD8f28n8EPTXDW5LF v9QveJRQIMp58gN0G1CzFhwRYjMB8Q=
9ldamu1tist5dbb9qtkntegrasihk0af.icann.org. 5 IN NSEC3 1 0 0 - 9LFI7DR6V1CMSVVD4KETK33CKBBLCHGD A NS SOA MX
TXT AAAA RRSIG DNSKEY NSEC3PARAM CAA

;; Query time: 47 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:36:48 PDT 2026
;; MSG SIZE rcvd: 381

(acw@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Kali - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Windows Server 2019 Kali

acw@kali: ~

```
Session Actions Edit View Help
TXT AAAA RRSIG DNSKEY NSEC3PARAM CAA

;; Query time: 47 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:36:48 PDT 2026
;; MSG SIZE rcvd: 381

(acw@kali)-[~]
$ dig +dnssec jonathan.icann.org.

; <<>> DiG 9.20.15-2-Debian <<>> +dnssec jonathan.icann.org.
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NXDOMAIN, id: 27961
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 0

;; QUESTION SECTION:
;jonathan.icann.org.          IN      A

;; AUTHORITY SECTION:
icann.org.                   5       IN      SOA     sns.dns.icann.org. noc.dns.icann.org. 2026040207 10800 3600
1209600 3600
icann.org.                   5       IN      RRSIG   SOA 13 2 3600 20260424091021 20260403014211 54015 icann.org.
j4E73oCUDysYlXoNVjWN5hNFJby+ErRZ4pskbzL9/WKBG0PK01BVEVvM ePPWYqG+Kke5Pdhx8HyvWqqsDYxliA=
9ldamu1tist5dbb9qtkntegrasihk0af.icann.org. 5 IN RRSIG NSEC3 13 3 3600 20260416004615 20260325231159 54015 i
cann.org. wJh2U17Id+GaUrWG+p5xgngwR/O/cFjy46v8uodhD8f28n8EPTXDW5LF v9QveJRQiMp58gN0G1CzFhwRYjMB8Q=
9ldamu1tist5dbb9qtkntegrasihk0af.icann.org. 5 IN NSEC3 1 0 0 - 9LFI7DR6V1CMSVVD4KETK33CKBBLCHGD A NS SOA MX
TXT AAAA RRSIG DNSKEY NSEC3PARAM CAA

;; Query time: 35 msec
;; SERVER: 172.16.36.2#53(172.16.36.2) (UDP)
;; WHEN: Thu Apr 02 21:37:12 PDT 2026
;; MSG SIZE rcvd: 386

(acw@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.